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Studierichting: Informatica
Oefeningenreeks 2B nr. 1.2

Druk $\sin(\text{Bgtan } x)$ uit in x :

$$\alpha = \text{Bgtan } x \text{ met } \alpha \in \left] -\frac{\pi}{2}, \frac{\pi}{2} \right[$$

$$\Leftrightarrow x = \tan \alpha$$

$$x^2 = \tan^2 \alpha$$

$$x^2 = \frac{\sin^2 \alpha}{\cos^2 \alpha}$$

$$x^2 + 1 = \frac{\sin^2 \alpha}{\cos^2 \alpha} + \frac{\cos^2 \alpha}{\cos^2 \alpha}$$

$$x^2 + 1 = \frac{\sin^2 \alpha + \cos^2 \alpha}{\cos^2 \alpha}$$

$$x^2 + 1 = \frac{1}{\cos^2 \alpha}$$

$$x^2 + 1 = \frac{1}{1 - \sin^2 \alpha}$$

$$\frac{1}{x^2 + 1} = 1 - \sin^2 \alpha$$

$$\sin^2 \alpha = 1 - \frac{1}{x^2 + 1}$$

$$\sin^2 \alpha = \frac{(x^2 + 1) - 1}{x^2 + 1}$$

$$\sin^2 \alpha = \frac{x^2}{x^2 + 1}$$

$$\sin \alpha = \frac{x}{\sqrt{x^2 + 1}}$$

$$\sin(\text{Bgtan } x) = \frac{x}{\sqrt{x^2 + 1}}$$

References